Quantum Physics

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1 Thermal radiation

1.1 Black Bodies

Definition: *Thermal Radiation* Emission of radiation of a body due to its temperature.

Definition: *Blackbody* A hypothetical perfect absorber and radiator of energy, with no reflecting power.

Definition: *Spectral Radiance* $R_\nu(T)$ Distribution of thermal radiation from blackbody. Power emitted per unit surface from an object at temperature T int the frequency interval $[v, \nu + d\nu]$. Integrating for all ν gives the *radiance*.
The *Spectral Energy Density* is the energy per volume at T in the interval $[\nu, \nu + d\nu]$. It is proportional to radiance $R(T) = \frac{c}{4}\rho(T)$.

Experimental results came with the *Stefan's Law*, that establish $R(T) = \sigma T^4$, where σ is the *Stefan-Boltzmann constant*.
As T increases, the frequency ν_{max} at which $R(T)$ is maximum also increases due to *Wien's Law* $\lambda_{max}T = 0.2898cmK$.

1.2 Planck's approach to radiation

Experimentally, Rayleigh have observed that for low frequencies $< E > \to kT$, but for high $< E > \to 0$. He proposed given the Boltzmann distribution $P_n = \dfrac{e^{-E_n/kT}}{\sum e^{-E_n/kT}}$:

$$E_n = nh\nu \implies < E >= \frac{\sum E_n P_n}{\sum P_n} = h\nu \sum nP_n = \frac{h\nu}{e^{h\nu/kT} - 1}$$

Integrating for each value we have:

$$\rho(T) = \frac{8\pi^5 k^4}{15c^3h^3}T^4 \implies R(T) = \sigma T^4 = \rho\frac{c}{4}$$

1.3 Photoelectric Effect

Einstein showed that given the electrons in a metal follow the following equation

$$eV_0 = K_{max} = h\nu - W_0$$

Where W_0 is the work function.

1.4 Relativity

We have to take it into account when velocities are close to c:

- $m = \gamma m_0$
- $E^2 = (pc)^2 + (m_0c^2)^2$
- $p = \frac{h}{\lambda}$ (*De Broigle relation*)

1.5 Compton Effect

1.6 X-Rays

Definition: *X-Rays* Waves with $\lambda = [0.01, 10]nm$
The energy of an X-ray is:

$$E = h\nu = K_0 - K_f \implies \lambda = \frac{hc}{K_0 - K_f}$$

2 Quantization and Atomic Models

2.1 Wave-Particle Duality

2.2 Uncertainty Principle

2.3 Atomic Models

2.3.1 Thomson

Model with negative electrons distributed in a continuous positive charge.
Atoms could be in two states:

- *Ground State* Electrons remains at equilibrium
- *Exited State* Electrons vibrate around equilibrium emitting electromagnetic radiation

2.3.2 Rutherford

Positive charge is concentrated in the nucleus, but some problems like fall of electrons arise with the model.

2.3.3 Bohr

Bohr's posutlates:

- The electron moves in a circular orbit around the nucleus due to Coulomb's Law
- The angular momentum of the orbit of an electron only can be $L = n\hbar$
- Electron does not emmit radiation
- Electron emmit radiation when hopping from orbit with frequency $\nu = \frac{E_i - E_f}{h}$

$$mvr = pr = n\hbar, \qquad p = \frac{h}{\lambda} \implies 2\pi \boxed{r = n\lambda}$$

Making equal the forces in the orbit

$$\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2} = m\frac{v^2}{r} \implies \boxed{r_n = \frac{4\pi\varepsilon_0 h^2}{mZe^2} \qquad n = 1, 2, 3, \ldots}$$

Velocity and Energy of orbits also quantized

$$E_n = -\frac{1}{2}\frac{Ze^2}{4\pi\varepsilon_0 r_n}\frac{m(Ze^2)^2}{2(4\pi\varepsilon_0\hbar)^2}\frac{1}{n^2} \implies \boxed{\nu_{i\to f} = \frac{m(Ze^2)^2}{\varepsilon_0^2(4\pi\hbar)^3}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)}$$

3 Schrödinger's Theory

4 One-electron Atoms

4.1 Previous considerations

Reduced mass
In order to take into account both forces ($F_{a\to b}$ and $F_{b\to a}$) we can work with:

$$\mu = \frac{Mm}{M+m} \implies F_e = \mu a_{rel}$$

Rydberg's constant

$$R_\infty = \left(\frac{1}{4\pi_0}\right)^2 \frac{me^4}{4\pi\hbar^3 c}, \qquad R_M = \frac{\mu}{m}R_\infty$$

4.2 TDSE in 3D

Developing the Schrodinger's equation with Coulomb potential $V = -\frac{Ze^2}{4\pi\epsilon_0 r}$:

$$-\frac{\hbar^2}{2m}\nabla^2\Psi - \frac{Ze^2}{4\pi\epsilon_0 r}\Psi = E\Psi$$

Suppose that $\Psi = R(r)\Theta(\theta)\Phi(\phi)$. Then we have the following ODEs:

Azimuthal

$$\frac{d^2\Phi}{d\phi^2} = -m_l^2\Phi \implies \boxed{\Phi = e^{im_l\phi}}, \qquad m_l = \ldots, -2, -1, 0, 1, 2, \ldots$$

Polar

$$-\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{m_l^2\Theta}{\sin^2\theta} = l(l+1)\Theta$$

Using the change of variable $z = cos\theta$:

$$\frac{d}{dz}\left((1-z^2)\frac{d\theta}{dz}\right) + \left(l(l+1) - \frac{m_l^2}{1-z^2}\right)\Theta = 0$$

The solution is given in terms of Legendre Polynomials P_l:

$$\boxed{\Theta(z) = (1-z^2)^{|m_l|/2}\frac{d^{|m_l|}P_l(z)}{dz^{|m_l|}}}$$

Radial

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{2\mu}{\hbar^2}(E - V(r))R = l(l+1)\frac{R}{r^2}$$

Using the parameters $\rho = 2\beta r, \quad \beta^2 = -\frac{2\mu E}{\hbar^2}, \quad \gamma = \frac{\mu Z e^2}{4\pi\epsilon_0\hbar\beta}$:

$$\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{dR}{d\rho}\right) + \left(-\frac{1}{4} - \frac{l(l+1)}{\rho^2} + \frac{\gamma}{\rho}\right)R = 0$$

If we consider ρ large we can express the solution in terms of Laguerre Polynomials F:

$$\boxed{R_{nl}(\rho) = F_{nl}(\rho)e^{-\rho/2}}$$

4.3 Interpretation

Note that Ψ depends of three parameters

- Principal $n = 1, 2, 3, \ldots$ (Total Energy)
- Azimuthal $l = 0, 1, 2, \ldots, n-1$
- Magnetic $m_l = 0, \pm1, \pm2 \ldots, \pm l$

4.4 Probability densities

Radial Probability density

$$\boxed{P_{nl}(r)dr = R_{nl}^*(r)R_{nl}(r)4\pi r^2 dr}$$

Angular Probability Density

$$\boxed{A_{nlm_l}(r)dr \propto \Theta^*(\theta)\Theta(\theta)d\theta}$$

Total Probability Density

$$\boxed{T = P(r)A(\theta)}$$

4.5 Angular Momentum and Spin

From the definition $\overline{L} = \overline{r} \times \overline{p}$ we get, as $\overline{p} = -i\hbar\nabla$:

$$L_x = -i\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \qquad L_y = -i\hbar\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \qquad L_z = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right),$$

We have also that ψ is eigenfunction of $L_z\psi = \hbar m_l \psi, \quad L^2\psi = \hbar^2 l\,(l+1)\,\psi$.

We write the magnetic dipole moment as:

$$\overline{\mu}_l = -\frac{eL}{2m} = -\frac{g_I \mu_b}{\hbar}\overline{L} \qquad g_I = 1$$

4.6 Magnetic Fields

If we introduce a magnetic field, the particle with magnetic moment u_l experiences a torque $\tau = u_l \times B$ and change of potential energy $\Delta E = -u_l \cdot B$

If the field is not uniform:

$$F_z = \frac{dB_z}{dz} u_l \hat{z}$$

The spin is a quantum number that does not come from the resolution of Schrödinger equation, but is added to match the theoretical developement with the experiments.
The value of the spin can only be $m_s = \pm\frac{1}{2}$

5 Elementary Particles

5.1 Two particles system

We will study a system with two particles. Then we will have a wave function $\Psi\left(\overline{r_1}, \overline{r_2}, t\right)$. This function must follow the Schrödinger's Equation.

Now we consider the potential $V(\overline{r_1}, \overline{r_2})$ time-independent. As we have shown in chapter 3:

$$\Psi\left(\overline{r_1}, \overline{r_2}\right)\phi(t) = \Psi(\overline{r_1}, \overline{r_2})e^{-\frac{iEt}{\hbar}}$$

But we cannot distinguish both particles if they are equal, but we can suppose $\Psi\left(\overline{r_1}, \overline{r_2}\right) = \Psi(\overline{r}_1)\Psi(\overline{r}_2)$

We define the symmetric S and antisymmetric A eigenfunctions as:

$$\begin{cases} \Psi_S = \frac{1}{\sqrt{2}}\left(\Psi_\alpha(\overline{r}_1)\Psi_\beta(\overline{r}_2) + \Psi_\alpha(\overline{r}_2)\Psi_\beta(\overline{r}_1)\right) \\ \Psi_S = \frac{1}{\sqrt{2}}\left(\Psi_\alpha(\overline{r}_1)\Psi_\beta(\overline{r}_2) - \Psi_\alpha(\overline{r}_2)\Psi_\beta(\overline{r}_1)\right) \end{cases}$$

In the case that they are confined in an infinite potential well of length a and $n_\alpha = 1, n_\beta = 2$ we have:

$$\left\{\Psi_\alpha = B\cos\left(\frac{\pi x}{a}\right)\Psi_\beta = \sin\left(\frac{2\pi x}{a}\right)\right.$$

5.2 Exclusion Principle

Weak Exclusion Principle: In a multi-electron atom, there can never be more than 1 electron in the same quantum state

Strong Exclusion Principle: A system containing several electrons must be described by an anti-symmetric eigenfunction